



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-04 Start Here

Storage, File Formats & Lakehouse - build trusted analytical table state above object files.

Artifact	C3-04
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



What changes in this phase

C2B gave you warehouse/layer/lakehouse mental models, and C3-01/C3-02/C3-03 gave you ingestion, transformation and orchestration. C3-04 now makes storage physical: file layout, partition/file health, table metadata, Delta commits, controlled schema change, MERGE, history and retention.

All 13 lessons are deeper engineering progression. Nothing here exists simply to reteach the Bridge.

Exact learner loop

- Study one 3-page C3-050 lesson.
- Reproduce its guided example with the lesson open.
- Close lesson and complete the matching Pxx in C3-051.
- Save code/config/output + confidence + main uncertainty + independent validation.
- Only then open the matching Rxx in C3-052.
- Close review and complete the changed retry.
- Mark Mastered only when fresh changed evidence validates.

Phase sequence

ID	Lesson
C3-04-L01	Object storage and namespace mental model
C3-04-L02	CSV, JSON, Avro and Parquet trade-offs
C3-04-L03	Columnar storage, statistics and compression
C3-04-L04	Partitioning strategy and pruning
C3-04-L05	Small-files problem and compaction
C3-04-L06	Catalogs, metadata and table namespaces
C3-04-L07	Lake vs warehouse vs lakehouse + open table-format trade-offs
C3-04-L08	Delta transaction log and ACID guarantees
C3-04-L09	Schema enforcement and schema evolution
C3-04-L10	MERGE, upsert, update and delete patterns
C3-04-L11	Time travel and reproducible historical reads
C3-04-L12	Medallion/layered architecture and table contracts
C3-04-L13	Optimization, retention and maintenance strategy



Assessment and runtime honesty

After all lesson mastery records, complete ColdChain. Then SolarFleet Gate A. A narrow weakness receives targeted repair + changed micro-retest; CampusRides/ClinicStock remain sealed until explicitly assigned as fresh full retests.

PASS requires every required competency to pass and zero unresolved critical failure. A failed required competency cannot be offset by stronger performance elsewhere.

Current local lane

PyArrow 25.0.1 and deltalake 1.6.3 are the local learner pins. Delta Lake project context is currently 4.4.0; local exercises use delta-rs because it is light enough for this phase. Iceberg 1.11.0 is used for architecture comparison, not as an additional mandatory runtime.

Mentor route contract

Today | Learn | Course | Repair | Evidence | Jobs | Reports | Settings. Each lesson gets an exact book page, practice page, review page and starter asset route.